# *Manip — cheat sheet*

Keyboard-driven algebraic manipulation. Lowercase letters, digits and symbols apply operations directly to the formula (the same keys work in the Enter menu); uppercase (Shift) are commands.

## NAVIGATION AND STEPS

| | |
|---|---|
| ← → ↑ ↓ | navigate |
| Ctrl ← → | move / take to the other side |
| Ctrl ↑ ↓ | leave / enter group |
| Shift ← → | select neighbors |
| Space | select |
| Esc | clear selection / leave |
| Enter | operations |
| Tab | workspaces |
| PgUp PgDn Home End | go through the steps |
| Ctrl Z | undo |
| Ctrl Shift Z | redo |

## COMMANDS (SHIFT)

| | |
|---|---|
| C | color |
| E | edit |
| Y | copy the expression (text) |
| L | copy LaTeX |
| P | present |
| K | keyboard |
| O | step note |
| R | record video |
| ] | larger formula |
| [ | smaller formula |
| \ | normal size |
| ? | shortcuts panel |

## ARITHMETIC

| | | |
|---|---|---|
| c | Calculate ↔ factor the number (12 ↔ $2^2 \cdot 3$) | $x = 7 - 3 \;\to\; x = 4$ |
| l | Combine like terms | $x = 2a + 3a \;\to\; x = 5a$ |
| x | Cancel | $x + 3 - 3 = 5 \;\to\; x = 5$ |
| 0 | Identity and zero: x + 0, 1·x → x; 0·x → 0; $x^0 \to 1$ | $x + 0 = y \;\to\; x = y$ |
| r | Sign rule: −(−a), (−a)b ↔ −ab | $x = a - (-b) \;\to\; x = a + b$ |
| - | Flip signs: A(B + C) → −A(−B − C) | $2(x - 1) = 4 \;\to\; -2(-x + 1) = 4$ |

## STRUCTURE

| | | |
|---|---|---|
| d | Distribute | $2(x+1)=4 \;\rightarrow\; 2x+2\cdot 1=4$ |
| e | Factor out | $1-a-b+ab=0 \;\rightarrow\; 1-a-b(1-a)=0$<br>with selection |
| ( | Wrap in parentheses | $a+b+c=0 \;\rightarrow\; a+(b+c)=0$<br>with selection |
| p | Remove parentheses | $a+(b+c)=0 \;\rightarrow\; a+b+c=0$ |
| g | Group selection | $a+b+c=0 \;\rightarrow\; a+(b+c)=0$<br>with selection |
| q | Add and subtract... | $x^2+6x=7 \;\rightarrow\; x^2+6x+9-9=7$<br>typing “9” |
| s | Substitute with... | $x+y=10 \;\rightarrow\; x+2x=10$<br>typing “2x” |

## FRACTIONS

| | | |
|---|---|---|
| j | Fractions: a/c + b/c ↔ (a + b)/c, (a/b) (c/d) ↔ ac/(bd) | $\frac{x}{2}+\frac{y}{2}=1 \;\rightarrow\; \frac{x+y}{2}=1$ |
| * | Multiply top and bottom by... | $x=\frac{1}{2} \;\rightarrow\; x=\frac{1\cdot 3}{2\cdot 3}$<br>typing “3” |

## POWERS

| | | |
|---|---|---|
| ^ | Powers and roots: aa ↔ $a^2$, $x^a x^b$ ↔ x^(a+b), $a^n b^n$ ↔ $(ab)^n$, √a√b ↔ √(ab), √x ↔ x^(1/2) | $x^2x^3=1 \;\rightarrow\; x^5=1$ |
| b | Binomial / multinomial ↔ special product | $(a+b)^3=0 \;\rightarrow\; a^3+3a^2b+3ab^2+b^3=0$ |

## FUNCTIONS

| | | |
|---|---|---|
| f | Log and exp: log(ab) ↔ log a + log b, log $a^n$ ↔ n log a, ln $e^x$ → x, change of base | $y=\log(ab) \;\rightarrow\; y=\log a+\log b$ |
| f | Trigonometry: $\sin^2+\cos^2=1$, tan = sin/cos, double angle, angle sum, parity | $y=\sin(2x) \;\rightarrow\; y=2\sin x\cos x$ |

Generated by Manip itself: the examples are the operations actually applied.

## COMPLEX NUMBERS

| | | |
|---|---|---|
| z | Conjugate properties | $w=\overline{ab} \;\rightarrow\; w=\overline{a}\,\overline{b}$ |
| ~ | z ↔ $\bar{\bar{z}}$ (double conjugate) | $w=\overline{\overline{z}} \;\rightarrow\; w=z$ |
| h | z·$\bar{z}$ ↔ $\lvert z\rvert^2$ | $w=z\bar{z} \;\rightarrow\; w=\lvert z\rvert^2$ |
| v | z ± $\bar{z}$ ↔ 2Re(z) / 2i·Im(z) | $w=z+\bar{z} \;\rightarrow\; w=2\,\mathrm{Re}(z)$ |
| y | Multiply by the conjugate | $w=\frac{1}{3+4\mathrm{i}} \;\rightarrow\; w=\frac{1(3-4\mathrm{i})}{(3+4\mathrm{i})(3-4\mathrm{i})}$ |

## EQUATION AND INEQUALITY

| | | |
|---|---|---|
| = | Apply to both sides... | $2x=4 \;\rightarrow\; \frac{2x}{2}=\frac{4}{2}$<br>typing “/ 2” |
| t | Swap sides | $x<4 \;\rightarrow\; 4>x$ |
| i | Estimate (≤, ≥, <, >)... | $y=x^2+1 \;\rightarrow\; y\geq 1$<br>typing “>= 1” |

## SYSTEMS

| | | |
|---|---|---|
| + | Add equations (select two) | $\begin{cases}x+y=3\\x-y=1\end{cases} \rightarrow \begin{cases}x+y=3\\x-y+x+y=1+3\end{cases}$<br>with selection |
| - | Subtract equations (select two) | $\begin{cases}x+y=3\\x-y=1\end{cases} \rightarrow \begin{cases}x+y=3\\x-y-(x+y)=1-3\end{cases}$<br>with selection |
| > | Substitute this variable into the other equations | $\begin{cases}y=2x+1\\x+y=7\end{cases} \rightarrow \begin{cases}y=2x+1\\x+(2x+1)=7\end{cases}$ |

## NAMES

| | | |
|---|---|---|
| n | Expand the name | |
| n | Name it... | $2(x+1)=y \;\rightarrow\; 2\boxed{K}=y$<br>typing “K” |
| w | Open in workspace | |